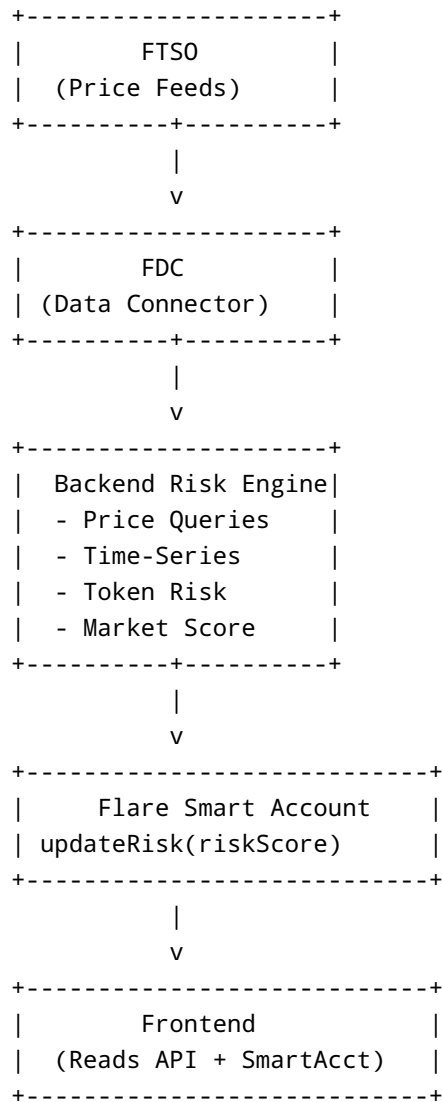


ARCHITECTURE.md

High-Level Architecture



Data Flow

1. **FTSO provides price feeds** used as the base data for all asset risk calculations.
2. **FDC (Flare Data Connector)** enables the backend to query reliable price and time-series data.
3. The **Backend Risk Engine** calculates **per-token risk scores** based on volatility, price deviation, and historical behavior.
4. The backend **aggregates all token risks** into one unified **market safety score**.
5. The backend **updates the Flare Smart Account** by invoking `updateRisk(score)`.

6. The **Frontend** fetches backend API data + reads the smart account state to visualize the real-time safety level.

Optional FAssets Story

- The backend also fetches **BTC** and **FAsset BTC** (e.g., FBTC) data.
 - It computes risk levels for each and also analyzes the **cross-chain difference**.
 - The frontend displays a **BTC vs FAsset BTC safety comparison** to highlight stability or divergence between chains.
-

FEATURES.md

Features Mapped to Judging Criteria

✓ Completion

- End-to-end functioning pipeline (FTSO → FDC → Backend → Smart Account → UI).
- Uses **all four Flare technologies** in a coherent workflow.

✓ Originality

- Unique **risk-scoring engine** built on top of real-time decentralized data.
- **Smart Account risk storage** turns market safety into an on-chain state.
- Tech used: **Smart Accounts, FDC, FTSO**.

✓ Learning

- Demonstrates deep understanding of:
 - **FTSO decentralized oracles**
 - **FDC structured data retrieval**
 - **FAssets cross-chain systems**
 - **Smart Account programmable logic**

✓ Design

- Clean UI showing:
 - Token-level risk
 - Aggregate market score
 - Smart account state indicator
 - BTC vs FAsset BTC comparison
- Tech used: **Smart Accounts + FAssets** for cross-chain visualization.

✓ Innovation

- Converts raw oracle data into a **market safety health index**.
- Cross-chain asset stability view using **FAssets + FDC + FTSO**.

- Integrates **smart account automation** to make scores on-chain.
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PITCH.md

30–60 Second Judge Pitch

FlareGuard is a real-time crypto safety engine built entirely on the Flare stack. We combine **all four major Flare technologies**—FTSO, FDC, FAssets, and Smart Accounts—into a single unified risk-intelligence product.

First, we use **FTSO** for decentralized price feeds. Then our backend pulls time-series data through **FDC** and computes risk scores for each token, plus an overall market safety score. We write that score on-chain using a **Flare Smart Account**, turning market conditions into a live blockchain state. And with **FAssets**, we compare native BTC to FAsset BTC to showcase cross-chain safety.

FlareGuard turns data into confidence—helping users instantly understand how safe the market is, across chains, powered by Flare.